

Paediatric ophthalmology and strabismus

Parts 1 and 2 | A paediatric intervention changes a developing system

Current anatomy and future visual development both matter.

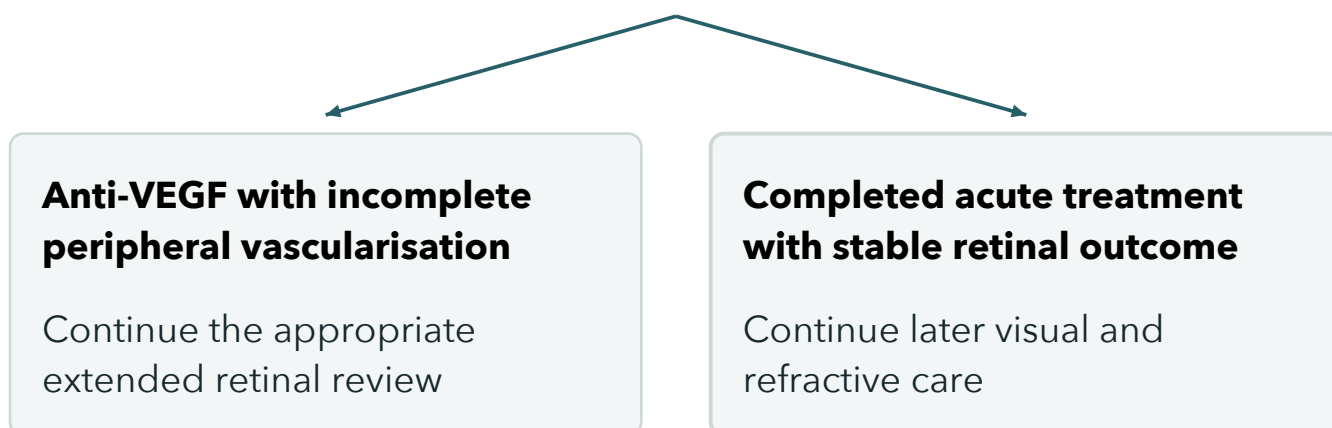
Finding	Implication
Premature retina	Record classification and vascular activity
Infant lens opacity	Plan focused image and deprivation prevention
Alignment disorder	Assess optics, fusion and mechanical pattern
Long-term plan	Identify the risk that outlasts the procedure

An infant's clear postoperative visual axis is only the first step: focused optical correction and amblyopia care are needed for development. ROP treatment similarly changes retinal activity without abolishing later refractive or alignment problems. In both settings, delayed complications can be asymptomatic; family support and dependable review are part of achieving useful vision.

ROP: treatment response changes but does not end review

Different modalities leave different surveillance obligations.

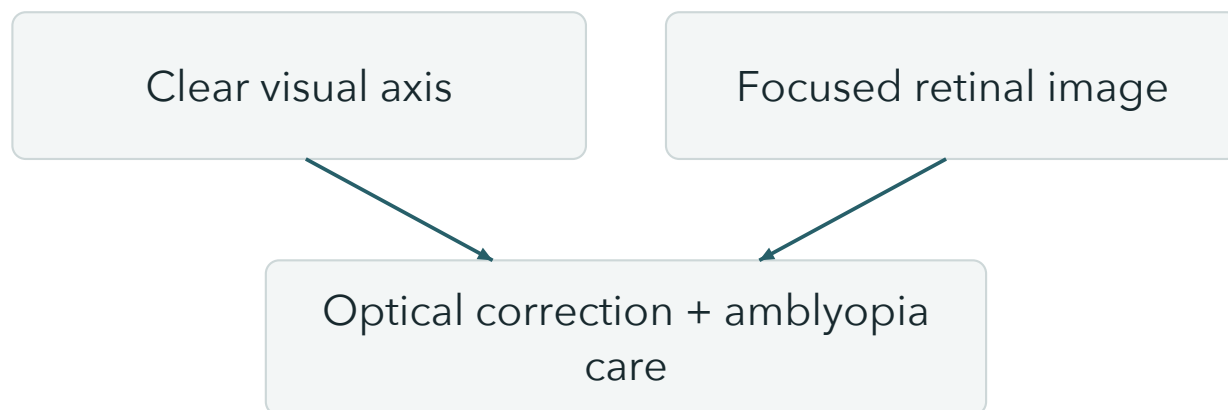
What remains after apparent regression?



After anti-VEGF treatment for ROP, visible regression can precede complete peripheral vascularisation and later reactivation. Transfer records should identify zone, stage, activity, treatment and the next retinal examination. The RAINBOW follow-up found less high myopia with ranibizumab than laser, but neonatal regression does not remove the child's later need for visual, refractive and alignment assessment.

Infant cataract: select optics and support adherence

An implant is one option, not an automatic developmental advantage.



Removing cataract is the start of visual rehabilitation, not its endpoint.

Original rehabilitation chain: clear axis, focused image and amblyopia care.

In the IATS young-infant unilateral cataract population, primary IOL implantation did not improve long-term acuity over contact-lens correction and increased early procedure burden. This does not settle every older-child decision. Aphakic refraction changes as the eye grows, requiring correction review; glaucoma risk continues years after surgery whether or not an IOL was implanted.

Strabismus: restore the relevant mechanism

Force direction, attachment and vascular reserve constrain the operation.

Finding	Implication
Altered muscle paths	Select geometry-directed correction
Major immediate underaction	Assess attachment failure promptly
Multiple rectus disinsertions	Consider perfusion risk and staging
Residual diplopia	Review useful gaze range and sensory context

When high myopia displaces the globe relative to extraocular muscles, restoring muscle-path geometry can matter more than a standard recession-resection dose. Profound new underaction immediately after surgery raises a slipped or lost muscle and warrants prompt assessment. Multiple rectus disinsertions reduce anterior ciliary supply, so vascular risk may favour staging or vessel-preserving alternatives.

Follow-up should have defined change triggers

Families need to know which new finding matters.

Finding	Implication
Optical correction	Growth and adherence require reassessment
Pressure and structure	Delayed risk can be asymptomatic
Retinal activity	Treatment-specific recurrence remains possible
Alignment and surface	Function can differ from an apparently good photograph

Infant cataract follow-up checks correction, amblyopia progress, pressure and ocular structure: useful acuity does not exclude delayed glaucoma. ROP surveillance separately asks whether activity has reactivated and whether vascularisation is adequate for the treatment pathway. New diplopia after muscle surgery and corneal exposure after lid surgery require targeted review rather than waiting for a routine annual visit.